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RECONSIDERING THE SIGNIFICANCE OF BIOGENETIC TIES

Introduction

Every other year, I teach an undergraduate course at North Carolina State University called "Religion, Gender, and Reproductive Technologies." The course uses assisted reproductive technologies as a lens to explore the internally diverse perspectives of Judaism, Christianity, and Islam. Various readings also introduce diverse feminist perspectives. The overarching goal of the course is to consider – with open minds – ways of responding to normative questions like what is a family, what is the nature of the relationship between parent and child, and what is a good life?

One of the main reproductive technologies we discuss is *in vitro* fertilization or IVF, where egg and sperm are combined outside of the body to create a human embryo. We also discuss the use of donor gametes (eggs and sperm). There are a few different scenarios in which a person might need a donated gamete to conceive a child. One would be if they are single (without a partner). Another would be if one member of a heterosexual couple has intractable infertility issues and cannot provide the needed egg or sperm. Finally, a same-sex couple would need a donated gamete to conceive. Each of these scenarios is distinct in practice. For example, a single woman need only find a sperm donor and undergo intrauterine insemination. A single man – or a gay male couple – needs both an egg donor and a gestational surrogate able and willing to undergo IVF and carry the pregnancy to term. Arguably, due to different risk levels and other considerations, these various scenarios raise distinct ethical questions.

The religious traditions we study all address the moral permissibility of donated gametes, or what they typically refer to as "third party" gamete donation. The assumption in the use of this terminology is that the gamete donor is an outside party to a married heterosexual couple. The Catholic Church is the most restrictive of the religious traditions we study, condemning both gamete donation and the process of IVF itself outright. But other religious traditions, even if they are accepting of IVF and deem gamete donation to be morally

permissible in at least some circumstances, nevertheless give reasons why relatedness between parent and child is important. None approaches gamete donation as if nothing is at stake morally if half a child's genetic inheritance comes from a stranger.

Thus, throughout the semester, we circle back again and again to the issue of biogenetic ties. Most often we encounter it in the context of discussing IVF with donated gametes, but it also arises in discussions of surrogacy and adoption. The inspiration for this essay grew out of these many conversations and my perception that biogenetic ties and their relationship to social ties between parents and children represent a core and underexplored bioethical issue in the 21st century.

The goal of this brief essay is to explore this question candidly: What *is* the moral significance of biogenetic relatedness between parent and child?[1] I suggest that in answering this question, the public conversation to date has insufficiently centered the perspectives of donor-conceived children, focusing instead on the needs and desires of prospective parents. But as more and more children born from these technologies come of age and articulate their own point of view, we have not only an opportunity to re-evaluate the meaning and importance of biogenetic ties, but perhaps an obligation to rethink our practices in the United States – where, for example, there is no restriction on how many times a man may donate his sperm, only a recommendation of 25 offspring per population of 800,000.[2] Half-siblings are finding each other online through websites such as the Donor Sibling Registry[3] and genetic ancestry registries like Ancestry.com. Some donor offspring are seeking to contact the donors who made their lives possible – for medical information, out of curiosity, or because they feel knowledge of the donor is an important part of their identity.[4]

Anyone who supports the use of reproductive technologies might question why the moral significance of biogenetic relatedness between parent and child requires further examination. To many minds, the biological connection between parent and child is self-evidently important and

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[1] This question is also addressed by Sally Haslanger, "Family, Ancestry, and Self: What is the Moral Significance of Biological Ties," in *Resisting Reality: Social Construction and Social Critique* (Oxford University Press, 2012), 158-182.
[2] The Practice Committee of the American Society for Reproductive Medicine and the Practice Committee for the Society for Assisted Reproductive Technology, "Gamete and Embryo Donation Guidance," *Fertility and Sterility* 122, no. 5 (November 2024): 804, https://doi.org/10.1016/j.fertnstert.2024.06.004.
[3] "The Donor Sibling Registry," accessed May 1, 2025, https://donorsiblingregistry.com/.
[4] See, for example, Michael Rothman, "Future People: The Family of Donor 5114," 2021, 1hr 37 min., https://futurepeoplemovie.com/.

natural. As such, its pursuit by adults desiring to be parents is both understandable and justifiable, using any available means. Indeed, the entire fertility industry exists to serve the human desire for biogenetic relatedness; otherwise, instead of IVF and its variants, we might see more people willing to adopt. We might even see the practice of adoption become better supported to meet the demand of adults who want to be parents.

Yet even if the strong desire for biogenetic relatedness is understandable, its pursuit raises potential concerns. Here I will elevate two. First, the pursuit of biogenetic connections (by any available means) risks treating the child as an object to which one has a right. Second, the pursuit of biogenetic connections (using donor gametes) risks reinforcing bionormativity – or the idea that a "true" family is a biologically related family, not a chosen family. In what follows, I elaborate these concerns and consider whether helpful insights for advancing the conversation might be gained from perspectives within Islam, Christianity, and feminist philosophy.

Does the Pursuit of A Biogenetic Connection Treat the Child as an Object To Which One Has A Right?

The right to procreate is generally thought to be a negative right, which means freedom from interference. A negative right is not an entitlement to positive assistance in procreating. It does not translate into entitlement to a child. In "Made, Not Begotten: IVF and the Right to Life Under Conditions," philosopher Susanne Kummer writes, "[N]o one should be prevented from founding a family; this is enshrined in international human rights law. However, this "right of defense" conversely/by implication does not give rise to a "right of claim," neither against the partner nor against the child."[5] The Reproductive Justice movement also defends the human right to *have* children and to raise them in a safe and healthy environment, in addition to the right *not* to have children.[6] These human rights frameworks accurately convey the importance of reproduction to human beings, regardless of their marital status, gender identity, or sexual orientation. But they do not imply the guarantee of a child.

To be clear, I am not questioning the human right to found a family. I am only suggesting that prioritizing biogenetic

[5] Susanne Kummer, Made, Not Begotten: IVF and the Right to Life Under Conditions, *The Linacre Quarterly* 89, no. 4 (2022): 424.
[6] Loretta Ross and Rickie Solinger, *Reproductive Justice: An Introduction* (University of California Press, 2017), 9-11.

relatedness can pit the interests of prospective parents and potential children against one another. Specifically, when using a donor gamete, an intended parent is satisfying their desire for a biogenetic connection to their offspring at the expense of the offspring's social connection to (or knowledge of) both of their biogenetic parents. In other words, the intended parent has decided that "half a loaf is better than none" when it comes to genetic connections: e.g., "You may not know your sperm donor father, but I know you are my biological child."

However, if the alternative to using a donated gamete to conceive a child is the child's non-existence, some argue the child has no grounds for complaint.[7] There is no other existence possible for this particular child, other than the one created with the unknown donor. This is a valid point, but I find this argument unsatisfying because it obscures the ways this practice may subordinate the emotional and psychological needs of the prospective child in favor of the emotional and psychological needs of the prospective parents. It may habituate us to not caring about the consequences for donor-conceived children.

As another source of cultural knowledge in a pluralistic society, religious traditions offer possible ways of evaluating the meaning of biogenetic relatedness between parent and child. For better or worse, religions offer centuries of wisdom, debate, and discernment. Their particular specification of a "good life" can be both a constraint and a source of guidance. In the context of a Religious Studies department in a secular university, our exploration is intellectual, not faith-based.

Many religious traditions address the importance of lineage in a family, such as Islam's concept of *nasab*. Clarity about one's lineage or family ties is important in Islam for a number of reasons. Anthropologist Marcia Inhorn writes:

> Islam is a religion that privileges – even mandates – biological descent and inheritance. Preserving the 'origins' of each child, meaning his or her relationships to a known biological mother and father, is considered not only an ideal in Islam but a moral imperative. In Islamic *fiqh* (jurisprudence), the tie by *nasab* (i.e., filiation, lineage, relations by blood) is considered to be one of God's great gifts to his worshippers. The preservation of *nasab* is emphasized through Qur'anic rules designed to ensure the sanctity of the family and the society; by preserving

[7] Melinda A. Roberts, "The Nonidentity Problem," *Stanford Encyclopedia of Philosophy*, Jul 19, 2024, https://plato.stanford.edu/entries/nonidentity-problem/.

> *nasab*, personal and social immorality are prevented, thus leading to the maintenance of society as a whole.[8]

In other words, biological ties are not incidental from an Islamic perspective; they do important work for instantiating a particular vision of a good society. The authoritative Sunni *fatwa* (the Al Azhar University *Fatwa* of 1980), which addresses the permissibility of IVF, unequivocally forbids the use donor gametes. In fact, it likens the use of donor gametes to adultery, or *zina*. The resulting child would be considered an illegitimate child of illicit sex. This judgment flows from the paramount importance of protecting unambiguous biogenetic ties – both for sake of the family and society, but also specifically for the sake of the child. In Inhorn's interpretation, "The problem with third-party donation…is that it destroys a child's *nasab* and violates the child's legal rights to known parentage, which is considered immoral, cruel, and unjust."[9]

Notably, the Al Azhar *fatwa* permits the use of IVF for married couples, provided there is no doubt that the couple is using their own gametes, meaning the husband's sperm and the wife's eggs. This support for IVF stems from the importance of preserving biogenetic ties and also a positive appraisal of science and medical technology more generally in Islam. As stated in the Al Azhar *fatwa*: "Prophet Muhammad, peace be upon him, mentioned the necessity to seek remedy for any disease, and sterility is a disease that might be curable; therefore to seek lawful treatment is then permissible."[10] A positive orientation to medical technology explains, in part, the enthusiastic embrace of ICSI (intracytoplasmic sperm injection), a variant of IVF that manually injects a single sperm into a retrieved egg. ICSI has allowed couples to overcome male-factor infertility while still maintaining a biogenetic connection with the social father. To accomplish ICSI, the wife must undergo IVF, with its associated risks, even if she herself has no medical issues impeding conception or pregnancy.

The authoritative Shia Muslim *fatwa* issued in 1999 by Ayatollah Ali Hussein al-Khamene'i also permits the use of IVF using the husband's sperm and wife's eggs. However, in striking contrast with the Sunni *fatwa*, the Shia *fatwa* also permits the use of donor gametes. The *fatwa* states: "It is legally not forbidden to fertilize a woman's egg with a donor sperm in and by itself, but the opposite gender [infertile man] should avoid touching or seeing the [female] child [naked], as these are

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[8] Marcia C. Inhorn, *The New Arab Man: Emergent Masculinities, Technologies, and Islam in the Middle East* (Princeton University Press, 2012), 233.
[9] Inhorn, *New Arab Man*, 234.
[10] The Al Azhar *Fatwa*, reproduced in Inhorn, *New Arab Man*, 327.

considered *haram*. In any case, a child born this way would not carry the name of his [social] father, but rather that of the sperm owner [the donor] and the egg and uterus carrier..."[11] The stipulation about the child's name suggests that preserving the child's known origins is of greater importance than satisfying the needs of the adults – which in this case might include avoiding the stigma of publicly acknowledging the need for a donor. In practice, the use of donor sperm with IVF has proven unappealing to many Muslim men precisely due to its disruption of a patrilineal connection.[12] The use of donor eggs has been somewhat more accepted among Shia Muslims. The lack of genetic connection between mother and child is thought to be at least somewhat compensated for by the biological connection of breastfeeding or 'milk kinship' (*rida'*), as well as pregnancy.

Christianity similarly contains a wide range of views about IVF and gamete donation. Protestant sects are the most diverse, ranging from openly supportive to highly critical. Speaking with a more unified voice, the Catholic Church defines a child's right to know his/her biogenetic heritage as central to respect for human dignity. The Church offers the following description of the child's rights, followed by what a child symbolizes for its parents:

"The child has the right to be conceived, carried in the womb, brought into the world and brought up within marriage: it is through the secure and recognized relationship to his own parents that the child can discover his own identity and achieve his own proper human development. The parents find in their child a confirmation and completion of their reciprocal self-giving: the child is the living image of their love, the permanent sign of their conjugal union, the living and indissoluble concrete expression of their paternity and maternity…"[13]

While sympathetic to couples who experience infertility, the Catholic Church deems the use of IVF to be morally impermissible under any circumstance, including when using the husband's sperm and wife's eggs. In the undergraduate course that I teach, students are generally aware of the

[11] The Fatwa of Ayatollah Ali Hussein Al-Khamene'i, reproduced in Inhorn, *New Arab Man*, 331.
[12] Soraya Tremayne, "The 'Down Side' of Gamete Donation: Challenging 'Happy Family' Rhetoric in Iran," in *Islam and Assisted Reproductive Technologies: Sunni and Shia Perspectives*, ed. Marcia C. Inhorn and Soraya Tremayne (Berghahn Books, 2012), 150-152.
[13] Congregation for the Doctrine of the Faith, *Instruction on Respect for Human Life in its Origins and on the Dignity of Procreation: Replies to Certain Questions of the Day*, 1987, https://www.vatican.va/roman_curia/congregations/cfaith/documents/rc_con_cfaith_doc_19870222_respect-for-human-life_en.html

Church's stance against abortion, but find themselves somewhat surprised by its rejection of IVF. Few know the reasons behind either teaching.

The reason for the Church's opposition to IVF is not that it opposes science or medical intervention. On the contrary, a surgical or pharmaceutical intervention that restored individuals' reproductive capacities and permitted natural conception to take place would be permissible. The problem is that IVF disrupts "procreative unity," or the belief that the unitive and procreative elements of a marriage should always be joined. Each and every act of sexual intercourse need not result in the conception of a child, but each should be open to it. Procreative unity lies behind the Church's objection to contraception. Likewise, the problem with IVF is that separates procreation from sex.

Because of its opposition to IVF using the husband's sperm and wife's eggs, it follows that the Catholic Church also opposes the use of donor gametes. But additional reasons are given for why seeking to preserve a partial biogenetic connection to a child (to one parent and not the other) is morally problematic. According to Church teachings, the adults, assumed here to be heterosexual and married, only have the right to become parents through each other. Having a child is not a personal project or a right:

"[M]arriage does not confer upon the spouses the right to have a child, but only the right to perform those natural acts which are *per se* ordered to procreation. *A true and proper right to a child would be contrary to the child's dignity and nature. The child is not an object to which one has a right, nor can he be considered as an object of ownership…"*[14]

Instead, the Church counsels couples experiencing infertility to adopt. The Catholic Church's enthusiastic embrace of adoption is itself an interesting contrast with Islam. Because of the importance of origin preservation, Islam supports something closer to permanent fostering rather than adoption (a full examination of which is beyond the scope of this paper).

These subtle and sometimes stark differences between religious traditions provide the occasion in our class discussions to try to discern underlying values and priorities. How is marriage being defined? Who is excluded? How are parents' perspectives valued? What role is gender playing? What concepts are used to describe what is owed to children or

[14] Ibid.

future generations? How is "*nasab*" different from the concept of "human dignity"? How is it similar? In addition, we always make a point of discussing how people of all faith traditions follow authoritative teachings in varying degrees. There are "official teachings," and there is "lived religion."[15] Crucially, we acknowledge that internal disagreements within faith traditions are complex and ongoing.[16]

Given this complexity, are there any helpful insights to be gleaned for advancing the conversation about the moral significance of biogenetic relatedness between parent and child? To be sure, there is no simple takeaway from the religious perspectives touched upon here. While we might appreciate the Catholic Church's centering of the rights of the child, for example, we might also be inclined to think the balance tips too far against the interests of prospective parents, especially given the Church's outright rejection of IVF, regardless of whose gametes are used, not to mention its position against abortion.

Likewise, we may appreciate the wisdom of the concept of *nasab* and respect the Islamic tradition's reverence for blood ties, but still question whose interests are being served by a stringent protection of origins: parents, specifically fathers, or children?[17]

What I see in these religious perspectives is not infallible judgment, but merely a substantive counterweight to the cultural and economic forces that fuel the fertility business. Religious perspectives offer fuller responses to the question of "why are biogenetic ties important?" They say that biogenetic ties are important not merely because prospective parents want them, or because prospective parents regard them as important. Rather biogenetic ties are important for the well-being children themselves, for families, and for the health of society.

Even so, the importance of biogenetic ties is not absolute.

Does the Use of Donor Gametes Challenge or Reinforce Bionormativity?

[15] Howard Brody and Arlene Macdonald, "Religion and Bioethics: Toward an Expanded Understanding," *Theoretical Medicine and Bioethics* 34, no. 2 (2013): 133–145.

[16] To take one example, Catholic theologians like Lisa Cahill have argued for the permissibility of IVF within marriage, using a couple's own gametes, by advocating for a more expansive, less literal interpretation of procreative unity.

[17] As mentioned above, anthropologist Marcia Inhorn writes in *The New Arab Man* that the advent of ICSI (intracytoplasmic sperm injection) has been a boon for many Middle Eastern couples experiencing severe male factor infertility. The genetic connection to the father is so important, it is worth putting a healthy wife through IVF.

The second concern I want to raise is whether the pursuit of biogenetic connections (by any available means) risks reinforcing bionormativity – or the idea that a "true" family is a biologically related family, not a chosen family. In "Family, Ancestry, and Self: What Is the Moral Significance of Biological Ties?" feminist philosopher Sally Haslanger – herself an adoptive parent – investigates the moral significance of biogenetic ties in the context of whether children have right to know their biological parents. She writes:

"The natural nuclear family schema plays an important role in forming identities – including healthy identities – in our current cultural context, and many people are stigmatized by not being able to "fit" the schema; in short, early twenty-first century American culture is bionormative. Being stigmatized is harmful and it is difficult to live a good life when stigmatized in this way. However, even granting the cultural significance of the natural nuclear family schema, there are two ways to combat the stigma. One is to provide resources so that everyone can come as close as possible to fitting the schema, another is to combat the dominance of the schema… I prefer the latter [strategy]."[18]

Rather than working to make it possible for all children to know their biological parents, e.g., by disallowing donor anonymity, Haslanger would rather work to make the dominant bionormative schema less dominant. Presumably, in that world, children's desires to know their biological parents would not be as strong. Biogenetic connections would matter less. Donor anonymity would be a non-issue, and the definition of what makes a family would be based on volition or social ties. Haslanger argues that just because something is common does not mean it is good, comparing the bionormative nuclear family to the subordination of women, which John Stuart Mill famously argued was unjust.[19] In making this comparison, Haslanger opens up the possibility of envisioning a future with different social norms.

With Haslanger, I agree that reducing the dominance of bionormativity would have real benefits, as any movement to promote greater acceptance and less judgment would benefit families. However, it is puzzling to say that bionormativity is reinforced by donor-conceived children who want to know their biological parents while at the same time permitting prospective parents to use anonymous donor gametes so that they, themselves, can preserve a biogenetic connection with

[18] Haslanger, "Family, Ancestry, and Self," 180.
[19] Ibid., 178.

their child. Here I would agree more with Olivia Schuman, who writes:

> [W]e should be cautious of the double standard of unfairly placing the burden of proof on donor-conceived individuals to prove the value of genetic knowledge, while not scrutinizing the motivations of parents who use anonymous gamete donors to help them meet social expectations and satisfy their own desires.[20]

Schuman is arguing against those who claim that a donor-conceived child's desire to know their genetic origins does not warrant accommodation. She is challenging those who would "den[y] the legitimacy of a right to know one's genetic origins." (145) Instead, Schuman grants the legitimacy of donor-conceived children's perspectives, a fuller picture of which is only recently emerging: "It is sometimes preferable to concede to the potential reinforcement of bionormativity when doing so is aligned with the presumed preferences of donor-conceived individuals."[21] One can appreciate efforts to normalize "nonbionormative" families while still recognizing and respecting the needs of donor-conceived children. Wishing for reduction of the bionormative schema will not make it so. In fact, allowing prospective parents to use anonymous gamete donors may reinforce bionormativity just as much as allowing for donor-conceived children to know their genetic origins.

I am by no means the first to suggest that the use of reproductive technologies to maintain a genetic connection to one or two parents reinforces the centrality of traditional, heteronormative families. In "(Queer) Family Values and 'Reciprocal IVF': What Difference Does Sexual Identity Make?," feminist philosopher, Amanda Roth argues that challenging bionormativity has historically been an important element of what she calls "queer family values," where "love makes a family." However, in her estimation, some forms of assisted reproduction seem to "buttress" rather than challenge bionormativity, inasmuch as they support "the more general notion that 'realness' is a matter of biology."[22]

Roth uses reciprocal IVF to frame her discussion. Reciprocal IVF (or R-IVF) is an assisted reproduction scenario in which one partner of a lesbian couple provides the egg and the other partner carries the pregnancy. The embryo is created with the

[20] Olivia Schuman, "Should Bionormativity Be a Concern in Gamete Donation?" *International Journal of Feminist Approaches to Bioethics* 16, no. 2 (2023): 154.
[21] Ibid., 143.
[22] Amanda Roth, "(Queer) Family Values and 'Reciprocal IVF': What Difference Does Sexual Identity Make?" *Kennedy Institute of Ethics Journal* 27, No. 3 (September 2017): 462, https://doi.org/10.1353/ken.2017.0034.

use of donor sperm. Roth ardently supports the right to use R-IVF while at the same time raising concerns about it:

> Similar to many positions that are skeptical of marriage as an institution but strongly favor the legalization of same-sex marriage, the case in favor of R-IVF is of a largely conditional sort: *If* analogous heterosexual ART is permissible and widely available, then R-IVF must be as well. Indeed, I accept that conclusion, but hold that the focus on legality, accessibility, and permissibility obscures consideration of further ethical issues: whether this is a choice-worthy option from the point of view of the queer community. I think there is good reason to question its choice-worthiness… though I hope it is possible to raise these concerns without condemning any individual choice to use R-IVF.[23]

In other words, while regarding as axiomatic the claim that LGBTQ parents are as legitimate as straight parents, Roth nevertheless provides reasons why same-sex couples should approach the use of reproductive technologies with a more skeptical eye.

I, too, am trying to walk that line: respecting individual desires for biogenetic connections while also throwing some skepticism at our rush to acquire them by any means necessary. It would seem cruel to discourage anyone from using reproductive technologies, as it seems to contradict a basic human desire for genetic connection. At the same time, I believe it is both hypocritical and disingenuous to deny donor-conceived children access to knowledge of their biogenetic parents in the name of discouraging bionormativity.

To subordinate the child's right to biogenetic ties in one's own pursuit of biogenetic ties is paradoxical and self-defeating.

Conclusion

What would it look like to balance the needs and interests of both prospective parents and future children, especially in a post-Dobbs world where reproductive rights have been so significantly curtailed? Restricting the use of IVF or ending the use of donor gametes is not the answer; these would be unnecessary overreactions. There are steps we could take in the United States that would honor the moral significance of biogenetic relatedness between parent and child without

[23] Ibid., 464.

making that significance absolute or predetermined by any particular worldview.

For one, donor-conceived people deserve to inhabit a world that has taken steps to pro-actively protect their interests. Eliminating donor anonymity while also supporting the ability of LGBTQ persons to access donor gametes could be one approach. Recently activists in the U.S. have pushed for an end to donor anonymity.[24] This advocacy is sometimes framed as troubling to LBGTQ families, "Some of these families fear that disclosure laws will open the door to recognizing biological donors in some way as parents — possibly granting them parental rights and more broadly undermining the legitimacy of L.G.B.T.Q. families."[25] But the elimination of donor anonymity need not be a threat if other protections are put in place to protect families formed through nongenetic bonds:

> For [some] L.G.B.T.Q groups, the solution isn't a return to secrecy but rather equal legal protection for families created through interpersonal bonds rather than genetic ones. They want states to pass the 2017 version of a model bill, the Uniform Parentage Act, which makes clear that a person who uses sperm or egg donation with the intent to be a parent is a parent, regardless of factors including genetic ties and marital status. The act also specifies that sperm and egg donors have no parental rights or responsibilities.[26]

This approach would both honor the significance of biogenetic connections *and* the equal moral status of families formed outside the "bionormative schema." Indeed, it might have the effect of reducing the dominance of bionormativity. Currently, only a few states have laws protecting families formed through nongenetic bonds. Thus, it would seem there is significant room for improvement in providing legal protections for both prospective parents and donor-conceived children.[27]

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[24] Emily Bazelon, "Why Anonymous Sperm Donation Is Over, and Why That Matters," *The New York Times*, Dec. 8, 2023, https://www.nytimes.com/2023/12/03/magazine/anonymous-sperm-donation-genetic-testing.html.
[25] Ibid.
[26] Ibid.
[27] Gary Nunn, "Why I Would Never Donate Sperm in the United States," *The New York Times*, Dec. 21, 2024, https://www.nytimes.com/2024/12/21/opinion/anonymous-sperm-donations.html. Nunn, a journalist based in Australia, writes, "U.S. laws ignore the psychological and emotional needs of children born through sperm donation. Before donating, I learned that many crave early contact with their donors to understand their identity, genetically inherited character traits and the personal connection of bloodline. Such relationships can form a crucial part of their sense of self and emotional well-being."

Another change would be to require fertility clinics to keep better records of donor-created offspring and their donors. Collecting this information would enable half-siblings to connect with each other if they desired. It would also help avoid half-siblings inadvertently dating, marrying, or having children together. Thanks to at-home genetic tests and the internet, it is possible – but by no means guaranteed – for donor-conceived children to find their half-siblings, or "diblings," on their own. While these relationships may or may not be significant to donor-conceived individuals, it is respectful (for a number of reasons) to facilitate this knowledge.

Finally, a related step could involve limiting the number offspring that may be created from a single donor. Rather than a professional society's unenforceable recommendation of 25 offspring per population of 800,000, the U.S. could adopt more uniform regulations akin to what is done in other countries. For example, Germany limits 15 offspring per donor. Britain allows a donor to donate to up to 10 families. China permits 5 pregnancies per donor.[28]

All of these suggestions seek to balance the interests of prospective parents and children. They consider the significance of biogenetic connections while also making space for the emergence of new norms.

[28] Ariana Eunjung Cha, "44 Siblings and Counting," *The Washington Post*, Sept. 12, 2018. https://www.washingtonpost.com/graphics/2018/health/44-donor-siblings-and-counting/.